

Machine Learning Algorithm Based Plant Root Contour Extraction in Edge Devices

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Abstract— The status of plant roots serves as a crucial metric reflecting the plant's growth state. Extracting and processing the contour of plant roots aids in analyzing the growth status effectively. In this study, high-resolution root images of plants were obtained using embedded optical imaging devices, transferred to edge computing nodes for image processing. By fine-tuning machine learning parameters, an artificial neural network model was developed to integrate Holistically-Nested Edge Detection (HED) and Canny edge detection algorithms.

This paper presents a methodology for model training and application based on convolutional neural networks on edge devices, typically single-board computers with limited memory (such as Raspberry Pi), addressing issues with traditional Canny edge detection's sensitivity to salt-and-pepper noise and susceptibility to producing false contours due to image gradient changes. The algorithm leverages HED, employing machine learning solutions through a large collection of plant root image data for pre-training the model, effectively filtering images while retaining accurate edge information. Furthermore, the output of this method serves as input for the Canny edge extraction algorithm, utilizing an adaptive thresholding algorithm to generate clear plant root contour images.

Experimental results demonstrate that the combined approach using embedded artificial neural network models and Canny contour extraction reduces device performance requirements while improving the quality of root contour extraction. This research contributes to a more accurate understanding of morphological features of plant root systems, providing comprehensive imaging data for plant root studies.

1. INTRODUCTION

The root system of a plant is one of the bases of its vital activities, and it plays an important role in plant growth and development, nutrient uptake, and water utilisation [1]. The morphological structure of the root system affects its distribution and absorption capacity in the soil. Also, the way the root system branches affects its nutrient uptake capacity in the soil. In addition, the root system also has the role of windbreak and sand fixation and the role of water conservation, which is also important for maintaining the balance and stability of the ecosystem. Therefore, analysing the root system of a plant will help to understand its growth and development, and edge extraction is an important means of obtaining the distribution of its root system.

At present, there are two main methods for edge detection in plant root system [2]: one is to choose the traditional image processing and computer vision methods, using the classical Robert, Sobel, Prewit and Canny algorithms; the second is through the artificial neural network and machine learning technology, based on a large amount of data training makes the construction of artificial neural network can be extract the edges of the plant root system to achieve accurate image analysis. Both methods take the colour image of the plant root system as input, and output the same width and height of the edge extraction image.

At present, scholars try to use three types of segmentation algorithms [3] threshold segmentation, edge detection and region growing in order to get the root image segmentation for different scenes.

On the basis of comprehensive comparison of traditional segmentation algorithms, many people have done exploration. Literature [4] introduces pulse-coupled neural networks which have a good performance in the field of image processing. By combining pulse-coupled neural network and Sobel operator, a segmentation method suitable for root images is proposed. The method is also compared with the traditional segmentation methods in terms of the amount of manual interaction, time consuming and mis-segmentation rate. The Zhang-Suen refinement algorithm is improved to obtain the morphology-related parameters of the root system. Literature [5] measured the three-dimensional (3D) coordinates and morphological parameters of the rice root system, and used to

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determine the initial position, growth direction and rate through statistical analysis of the data. Literature [6] used ground-penetrating radar to scan the plant root system, extracted the spatial coordinates of the roots from the detection results, and used BP neural networks to estimate the diameter of the root points. Literature [7] uses an endoscopic device to collect pictures of the root system and uses the wavelet transform to process and detect the root pictures.

Currently, the more classical edge detection operators are Robert, Sobel, Prewitt and Canny operators [2], the advantage of applying these algorithms for image edge detection is that the computation is less complex, however, the effect of the image noise is also larger, thus these algorithms are not very satisfactory in making edge detection. Classical algorithms tend to construct a neighbourhood centred on the pixel to be processed as the basis for grey scale analysis to achieve the extraction of the image edge profile [8]. However, such methods have several drawbacks: firstly, due to the influence of noise, it leads to inaccurate edge positioning; secondly, it is easy to bring the defect of edge blurring or even misclassification.

Therefore there have been many attempts to use artificial neural networks for edge detection, and the application of deep learning methods has become a watershed in the field of edge detection [2]. One of the more pioneering works is an edge detection algorithm called holistically-nested edge detection (HED) proposed by Saining Xie and Zhuowen Tu et al. in 2015 [9], which utilises deep learning techniques to automatically learn rich hierarchical features, significantly improving the performance of edge and object boundary detection. The algorithm belongs to the deep learning algorithms with end-to-end multi-scale models, cleverly utilising adaptations of VGGNet-16, and its model is simple enough to be applicable to the extraction task with the simple feature of plant root distribution. Therefore, in this paper, we will adopt the HED network model for edge detection of plant root system and try to use it in combination with traditional operators.

This paper consists of the following sections, the first section about the introduction of this study, the second section is an overview of the HED principle, the third section is about the parameter tuning and scheme selection, the fourth section is about the analysis of the experimental results and the last subsection is about the conclusion.

2. HOLISTICALLY-NESTED EDGE DETECTION PRINCIPLE

The HED algorithm addresses important problems in the field of computer vision, including global image training and prediction, and nested multi-scale feature learning. Compared with other edge detection algorithms based on convolutional neural networks, the HED algorithm not only achieves significant improvements in performance, but also has a faster speed.

The features of the HED algorithm include global image training and prediction, and nested multiscale feature learning. Global image training and prediction means that the system takes images as input and directly generates edge images as output, thus achieving end-to-end image prediction, as shown in the neural network block diagram1. Nested multiscale feature learning performs deep multiscale feature learning by “guiding” early classification results through deep layers of supervision. This feature allows the HED algorithm to automatically learn rich hierarchical features, thus better solving the challenging problems in edge and object boundary detection.

Additionally, the HED algorithm utilises the techniques of Fully Convolutional Neural Networks (FCN) and Deeply Supervised Networks (DSN) [10]. FCN-like image-to-image training allows the system to be trained on a large number of samples at the same time, which improves performance. Deep supervised networks lead to more transparent feature learning, enabling the system to better understand image features.

In addition, the HED algorithm encourages coherent contributions from each layer through side-output interpolation in end-to-end learning, which improves performance. Taken together, the HED algorithm achieves automatic learning of rich hierarchical features through features such as global image training and prediction, nested multi-scale feature learning, and utilisation of FCN and DSN techniques, which significantly improves the performance of edge and object boundary detection with faster speed. This makes the HED algorithm an important breakthrough in the field of computer vision, providing a new solution to the edge detection problem.

3. PARAMETER TUNING AND OPTIMISATION PROCESS AND PROGRAMME SELECTION

In this experiment, the trained model can be seamlessly tailored and ported to embedded devices. The employed paddle-lite enables artificial neural networks to run on resource-constrained edge devices [11] to perform artificial intelligence tasks such as image processing, speech processing,

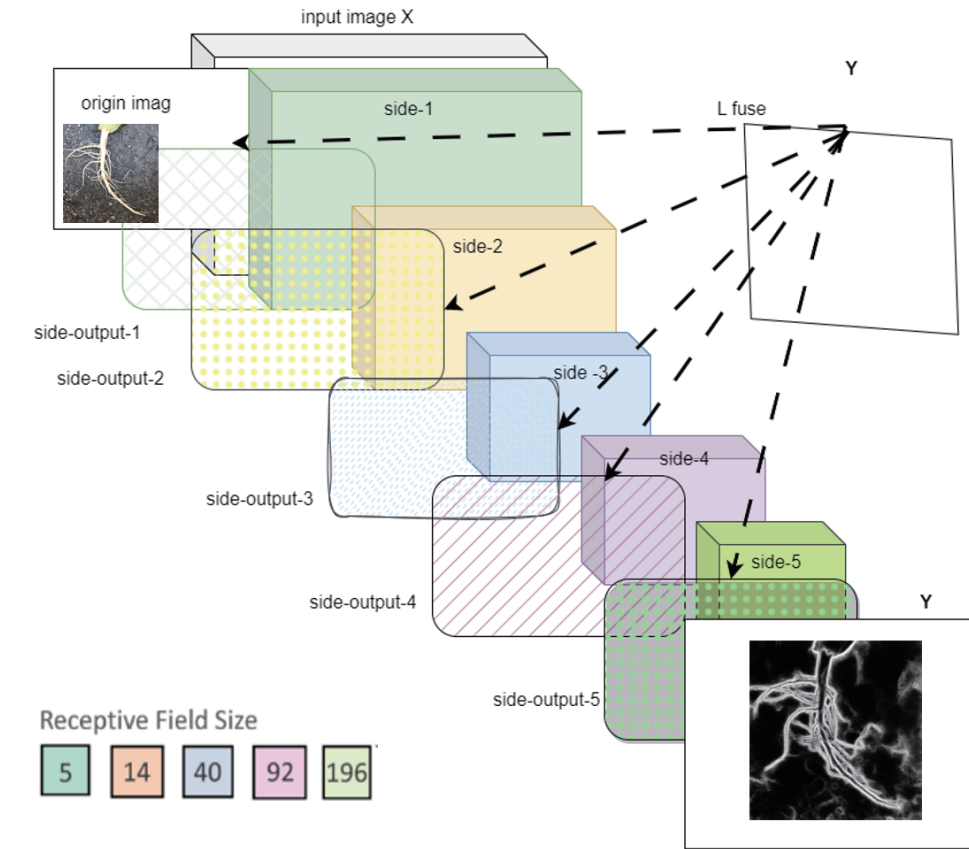


Figure 1: HED overall nested edge detection model framework.

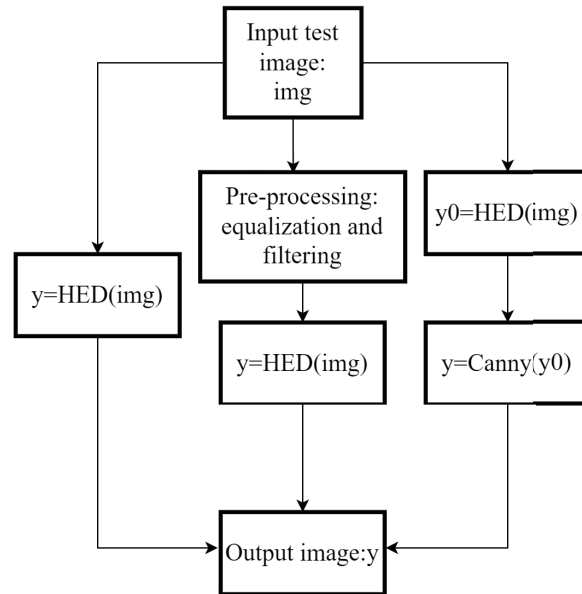


Figure 2: Programme selection.

etc.. In this study, the CPU version of paddle is used and is trained and tuned based on publicly available edge image segmentation datasets, in addition to a plant root contour dataset (20 sheets in total) obtained using script processing for parameter tuning. The parameters of the model from the original HED paper were tuned to achieve its ability to generalise to contour detection of plant root images and enable a usable state to be reached.

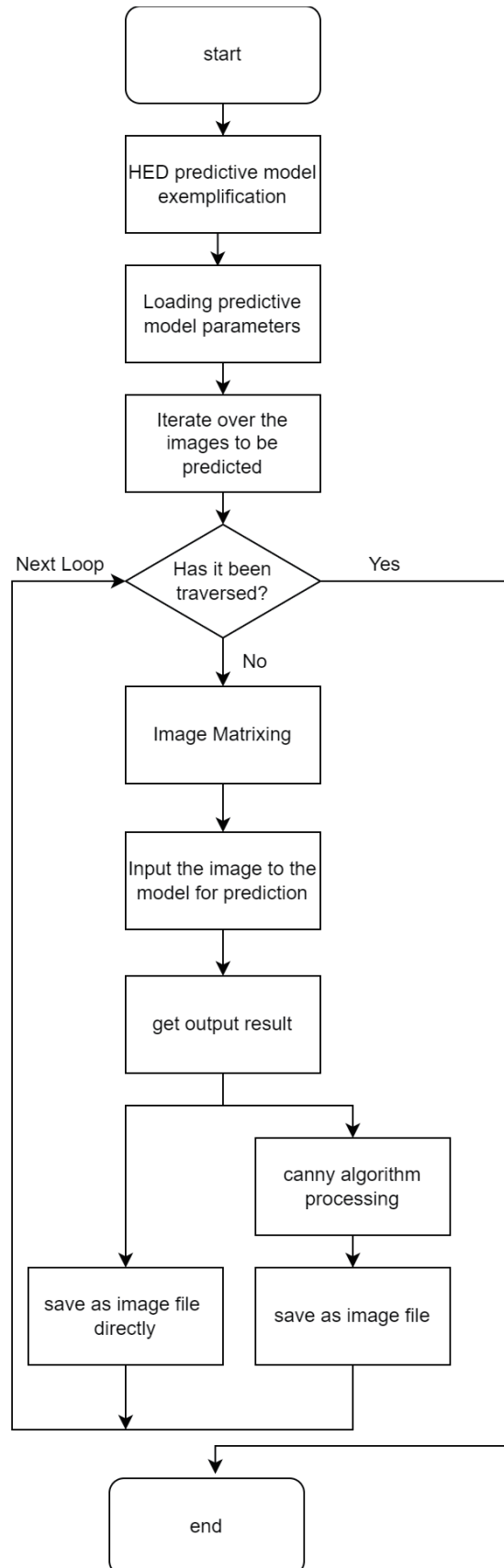


Figure 3: HED model prediction flowchart.

After successfully training the HED model, as shown in Figure 2, there can be three options for applying this HED model, the first is to use this model directly for prediction to reason about the edge contours of the plant root system. The second is to carry out greyscaling of histogram equalisation and time-frequency domain filtering before calling the model, and inference is carried out after inputting it into the model, and the third is to carry out a step of conventional edge detection after obtaining the inferred graph, using either the sobel operator or the canny operator, choosing the appropriate threshold value, and finally obtaining the final result.

After comparing the advantages and disadvantages of these three options in this study, the process of calling the model first and then using the Canny operator for processing was chosen. To use the HED model to process plant root images, it is necessary to ensure that the information of the plant root images is more complete. If a preprocessing scheme is adopted, according to the Information Processing Theorem, a part of the image information has already been lost during processing, which affects the prediction process of the HED model. And directly call the model, may wish to join the post-processing process, because the output of the call model can be saved separately and then post-processing and save the post-processing image, one can directly get the results predicted by the HED model, but also and Canny post-processing to do the results of the comparison, to choose the optimal image to save.

Therefore, the detailed workflow of this study is shown in the following Figure 3, which also includes the workflow of the prediction script for root contour extraction using the HED model. Firstly, the model network is exemplified and then loaded with the model parameters obtained from the training, then all the images in the directory of images to be predicted are traversed and each image to be predicted is matrixed and then fed into the HED model for prediction, once the result is predicted, it is firstly converted into image format and saved into the directory of the prediction results, and then the prediction is continued to be carried out again using Canny operator for the prediction results. Then continue to use the Canny operator to carry out another edge extraction on the prediction results, the threshold value can be reasonably selected according to the image characteristics, and finally also converted into image format and saved in the Canny detection results directory.

4. EXPERIMENTS AND ANALYSIS OF RESULTS

It should be prepared a number of plant root images and placed them in the test folder, and then, created the HED model prediction results folder, the original image Canny edge extraction folder, and the HED prediction results Canny edge extraction folder. And put the prediction script and

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[INFO] Paddle Version:2.4.2
[INFO] Loading pretrained model from model_hed.pdparams
[INFO] There are 38/38 variables loaded into HED.
[INFO] test_path D:/WorkSpace/plant_root/plat_root_pic/ori/
[INFO] target_path D:/WorkSpace/plant_root/plat_root_pic/hed_res
[INFO] target save path is already, your option would rewrite the dir!!!
[INFO] hed-canny target save path D:/WorkSpace/plant_root/plat_root_pic/hed_canny
[INFO] ori-canny target save path D:/WorkSpace/plant_root/plat_root_pic/ori_canny
totally 20 images
[INFO] Saving 'D:/WorkSpace/plant_root/plat_root_pic/ori01', Processing 1 of 20...
[INFO] Saving 'D:/WorkSpace/plant_root/plat_root_pic/ori02', Processing 2 of 20...
[INFO] Saving 'D:/WorkSpace/plant_root/plat_root_pic/ori03', Processing 3 of 20...
[INFO] Saving 'D:/WorkSpace/plant_root/plat_root_pic/ori04', Processing 4 of 20...
[INFO] Saving 'D:/WorkSpace/plant_root/plat_root_pic/ori05', Processing 5 of 20...
[INFO] Saving 'D:/WorkSpace/plant_root/plat_root_pic/ori06', Processing 6 of 20...
[INFO] Saving 'D:/WorkSpace/plant_root/plat_root_pic/ori07', Processing 7 of 20...
[INFO] Saving 'D:/WorkSpace/plant_root/plat_root_pic/ori08', Processing 8 of 20...
[INFO] Saving 'D:/WorkSpace/plant_root/plat_root_pic/ori09', Processing 9 of 20...
[INFO] Saving 'D:/WorkSpace/plant_root/plat_root_pic/ori10', Processing 10 of 20...
[INFO] Saving 'D:/WorkSpace/plant_root/plat_root_pic/ori11', Processing 11 of 20...
[INFO] Saving 'D:/WorkSpace/plant_root/plat_root_pic/ori12', Processing 12 of 20...
[INFO] Saving 'D:/WorkSpace/plant_root/plat_root_pic/ori13', Processing 13 of 20...
[INFO] Saving 'D:/WorkSpace/plant_root/plat_root_pic/ori14', Processing 14 of 20...
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[INFO] Saving 'D:/WorkSpace/plant_root/plat_root_pic/ori19', Processing 19 of 20...
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=====end=====

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Figure 4: Output results of running the prediction script.

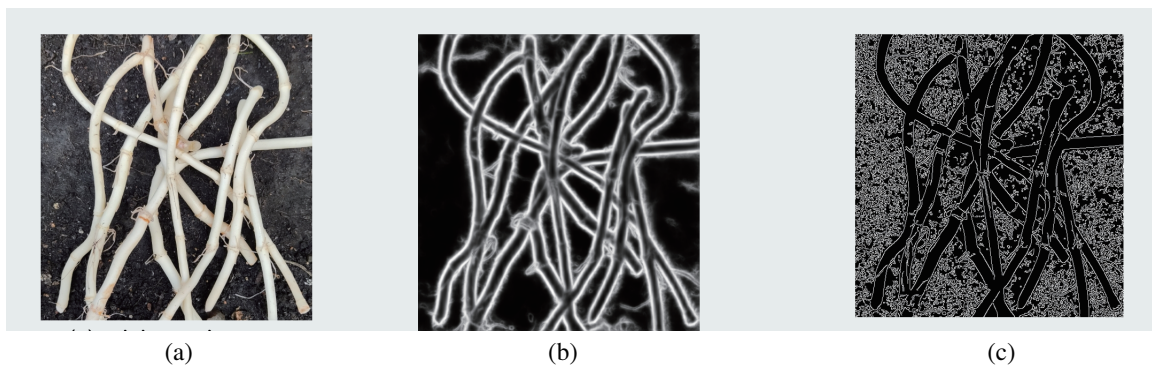


Figure 5: HED model detection results comparison Figure 1. (a) Origin raw image. (b) HED output. (c) Canny output.

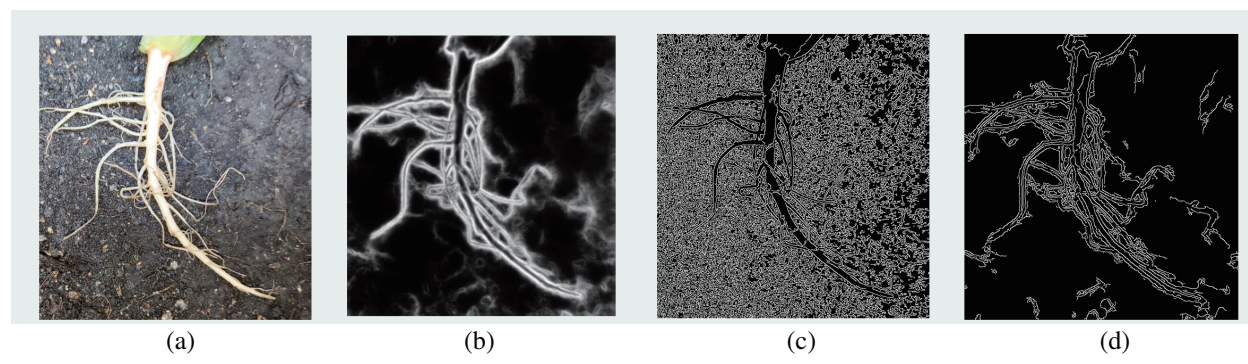


Figure 6: HED model detection results comparison Figure 2. (a) raw image. (b) HED output. (c) Canny output from a. (d) Canny output from b.

prediction model into the workspace folder, start the python prediction script to get the result output. Figure 4 shows the printed information of the prediction script at the output screen.

As shown in Figure 5, the original image, the results of the HED model for predicting the plant root system, and the results of Canny's edge extraction of the original image are given respectively. Figure 5(a) is the original figure. Figure 5(c) Edge detection using the traditional Canny operator still results in many pseudo-edges being detected. In Figure 5(b), the main trunk of the plant's root whiskers is extracted, and the fine gravel in the soil is also well removed. It can be seen that the edge extraction using the overall nested artificial neural network is more accurate than the traditional Canny operator edge extraction, but the HED model is not good at image brightness and darkness uniformity.

The results of the joint post-processing using the HED model are shown in Figure 6, Figure 6(a) is the original image, Figure 6(b) is the result after the prediction of the HED model, and the main edges of the plant are clearly contoured, and the impurities in the soil are filtered out. Figure 6(c) is the result of the original image after edge extraction by Canny operator, there are still many pseudo edges and noise caused by soil. Figure 11(d) is the result of the HED model prediction result after Canny operator edge extraction again, with less noise than Figure 6(c), showing the main root whiskers, and also with pseudo edges.

5. CONCLUSION

In summary, it can be seen that the using of HED model for plant root prediction can well circumvent the noise problem of the traditional canny operator, although it introduces the problem of uneven contour brightness and darkness, but the root contour recognition is more accurate, the probability of pseudo contour is low, it is an excellent plant root contour extraction scheme. Some shortcomings in the experiments are the high number of training parameters, which consumes time to train the model, as well as the low level of trimming of the model.

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